Campus Pond Water Survey

Environmental Science Club
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Abstract

The campus pond is home to fish, birds and many plants, and it collects rain water from the roofs of three faculty buildings. Over four weeks in May 2026 the club measured the temperature, acidity and oxygen level of the water at three points around the pond. This report describes how the samples were taken, presents the results, and suggests simple steps that could keep the pond healthy.

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Chapter 1

Introduction

1.1 Background

The pond was dug in 1998 to hold storm water during the rainy season. Since then it has become a favourite place for students to sit between classes, and a small ecosystem has grown around it.

Nobody has measured the quality of the water for several years. After a period of hot weather in April, some members noticed green water near the outlet and a few dead fish, so the club decided to run a short survey.

1.2 Aims

The survey had three aims.

1. Measure basic water quality at fixed points every week.
2. Find out whether the quality differs around the pond.
3. Suggest low-cost actions for the grounds team.

Chapter 2

Methods

2.1 Sampling Points

Three sampling points were marked with wooden posts: the inlet pipe on the north side, the footbridge in the middle, and the outlet on the south side. Samples were taken every Monday at 9 a.m., about 20 cm below the surface.

2.2 Measurements

2.2.1 Temperature and pH

A digital thermometer and a pocket pH meter were used at each point. The pH meter was calibrated before every visit.

2.2.2 Dissolved Oxygen

Dissolved oxygen was measured with a test kit that changes colour. Each reading was repeated twice and the average was recorded.

Chapter 3

Results and Discussion

3.1 Temperature and pH

The water temperature rose slowly from 29 to 32 degrees during the month. The pH stayed between 7.0 and 7.8 at all three points, which is a normal range for a pond with plants.

3.2 Dissolved Oxygen

Oxygen levels were good at the inlet and the footbridge, but they fell below 4 mg/L at the outlet in the last two weeks. Water near the outlet hardly moves, and dense algae grow there in hot weather.

3.3 Suggestions

- Remove some of the floating plants near the outlet every month.
- Add a small solar fountain to keep the water moving.
- Repeat the survey at the end of the rainy season.

Acknowledgements

We thank the Faculty of Science and Technology for lending the test equipment, and the grounds team for their help at the outlet.

Appendix A

Field Checklist

Members took this checklist to the pond on every visit.

1. Thermometer, pH meter and oxygen test kit
2. Three clean sample bottles, labelled A, B and C
3. Notebook, pencil and a watch
4. Gloves and a towel